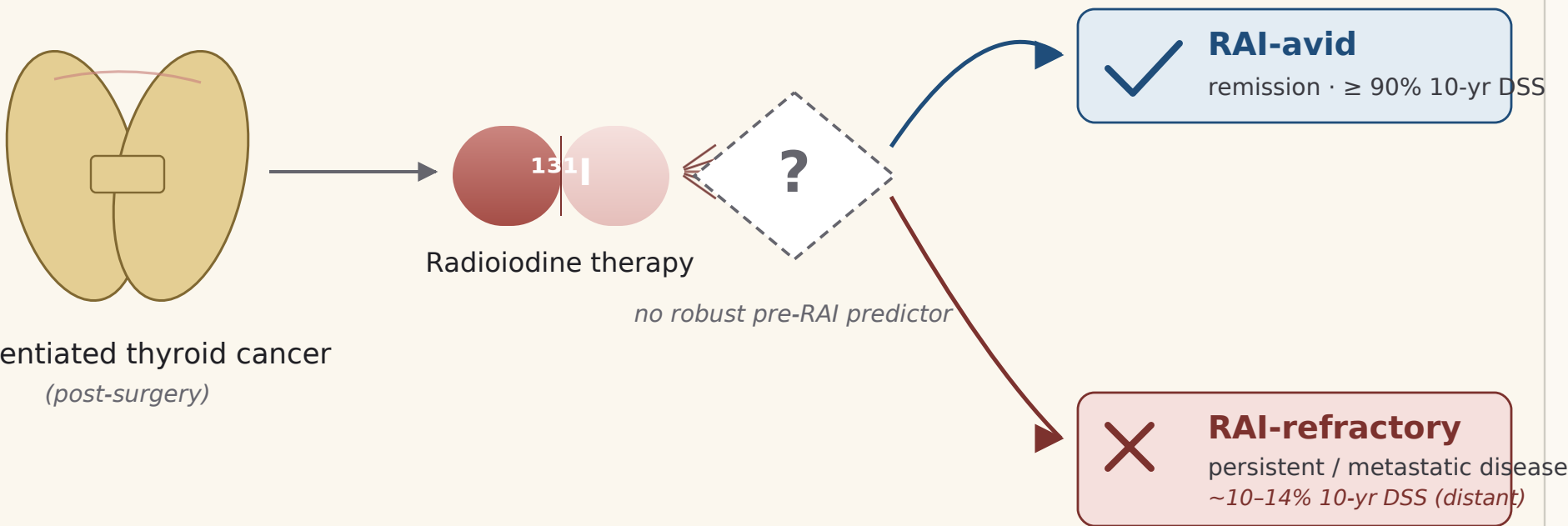


A multi-gate model of radioiodine failure in thyroid cancer

An eight-gene differentiation / iodide-handling axis linking discovery to RAI-labelled validation cohorts

a Clinical unmet need

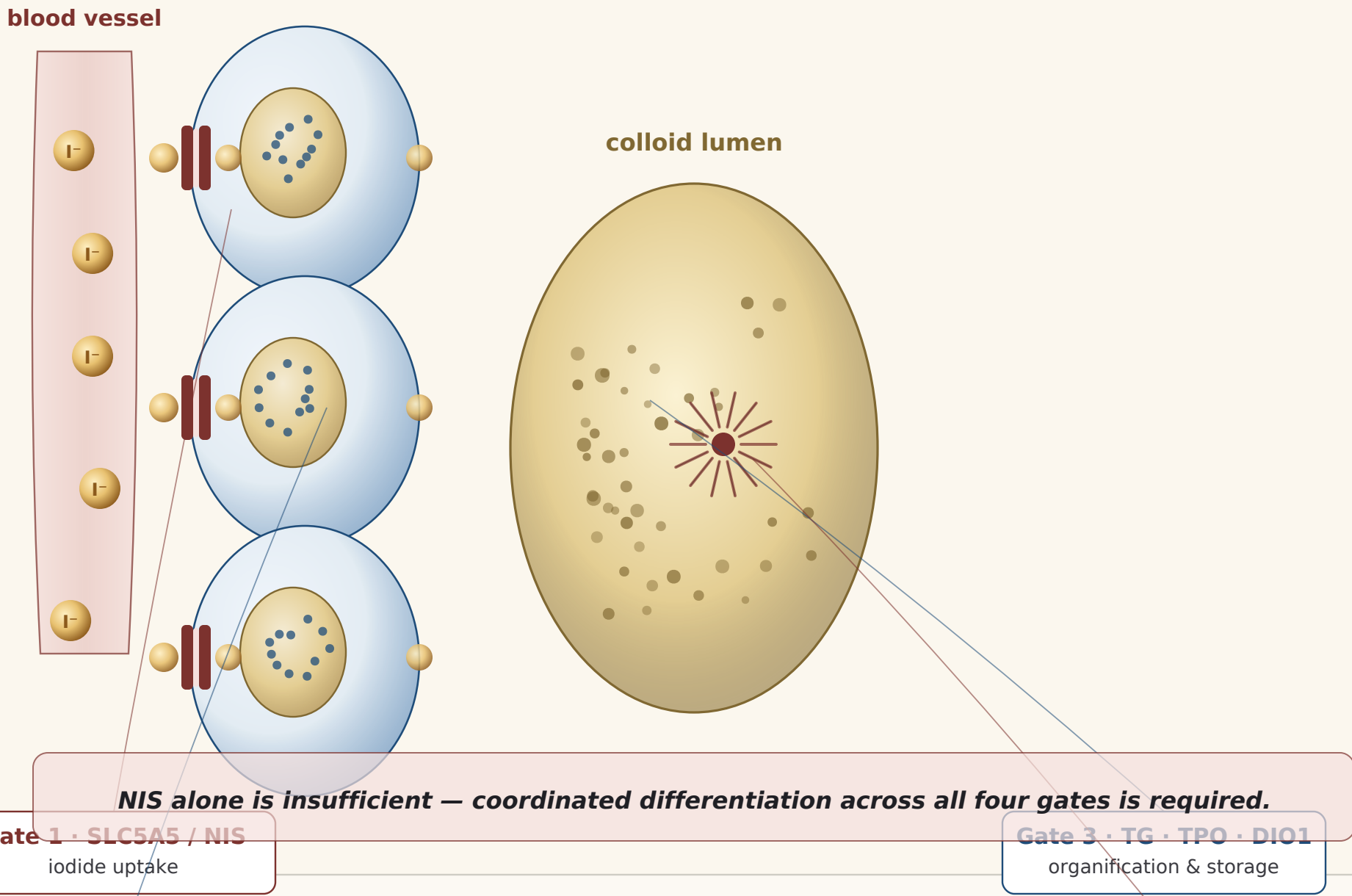


Need: a parsimonious molecular readout to triage RAI candidates

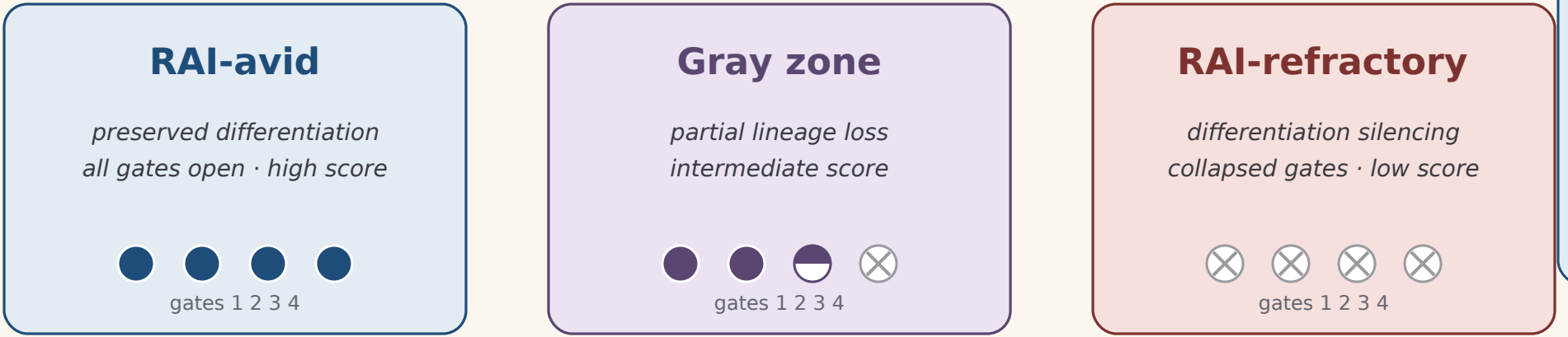
Current biomarkers (BRAF · RAS · TERT) do not reliably partition the RAI-refractory population.

b

Multi-gate biology of RAI response



c Three molecular states along the eight-gene differentiation-silencing axis



8-gene differentiation-silencing axis

preserved silenced

SLC5A5 · TPO · TG · TSHR · PAX8 · NKX2-1 · FOXE1 · DIO1

BRAF

RAS

TERT

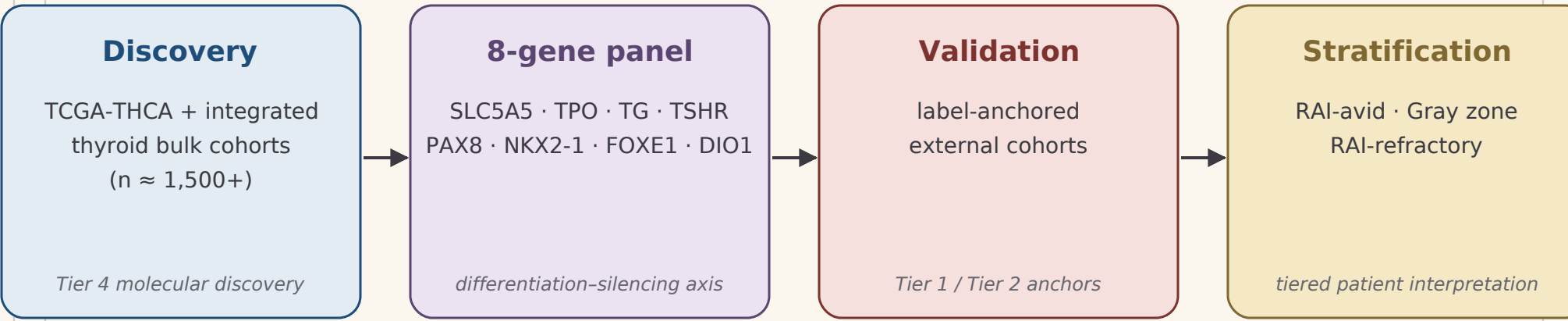
TP53

Fusion

genomic modifiers · secondary, not the main axis

d

Tiered evidence ladder · discovery → validation → stratification



GSE151179

RAI avidity

Tier 2

GSE299988

supportive

Tier 2

Boucai 2023

RECIST exceptional

Tier 1

Mu 2024

4-class uptake

Tier 2 · HRA004166

Tier-4 discovery → parsimonious 8-gene panel → label-anchored validation → three-class stratification

Positioned as a risk-stratification readout — prospective Tier-1 confirmation is the next step.